

“Federal Contract Oversight: FY2025 Risk Classification Framework”

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. ABSTRACT

The federal government awards over \$700 billion in contracts annually, yet no standardized automated framework exists for identifying high-risk awards at scale. This paper presents an end-to-end machine learning pipeline for federal contract risk detection using FY2025 transaction data from USASpending.gov, covering 600,862 contracts across the Department of Defense (DoD), Department of Homeland Security (DHS), and Department of Health and Human Services (HHS). In the absence of labeled training data, a novel three-algorithm majority vote framework combining K-Means clustering, Isolation Forest, and Hierarchical Density-Based Spatial Clustering of Applications with Noise (HDBSCAN) was developed to construct defensible ground truth labels. Supervised classifiers - Logistic Regression, Random Forest, and XGBoost - were then trained on 56 engineered risk features spanning six dimensions. XGBoost with balanced class weights achieved the best performance, with a cross-validated ROC AUC of 0.7574 (± 0.003) and recall of 59.1%. SHapley Additive explanations (SHAP) analysis identified contract financing type as the dominant risk driver. The system flagged 39,161 contracts (8.4%) as high risk, providing a traceable ranked list for oversight review.

Keywords - *federal procurement, anomaly detection, unsupervised learning, XGBoost, SHAP, contract risk, USASpending.gov, HDBSCAN, Isolation Forest, K-Means.*

I. INTRODUCTION

A. Problem Definition

The United States federal government obligates more than \$700 billion in contracts each fiscal year across hundreds of agencies and thousands of vendors. The sheer volume of transactions - exceeding 600,000 contract actions annually - makes comprehensive manual review by acquisition officers and auditors infeasible. The Government Accountability Office (GAO) has repeatedly identified federal contracting as a high-risk area susceptible to fraud, waste, and abuse [1]. Despite this designation, oversight efforts remain reactive, relying on post-award audits and sampling methodologies that cover a small fraction of total contract activity.

Three structural challenges compound this problem. First, no standardized risk taxonomy exists across agencies. Second, no publicly available human-annotated dataset of federal contract risk labels has been published, precluding direct supervised learning. Third, risk in federal

procurement is inherently multi-dimensional, encompassing financial irregularities, competition anomalies, vendor characteristics, compliance gaps, performance duration, and socioeconomic misrepresentation.

This project addresses these gaps by constructing an automated, data-driven pipeline that assigns risk classifications to federal contracts at scale, providing oversight teams with a prioritized list of awards warranting further review.

B. Background Information

Federal procurement oversight has historically relied on two primary mechanisms: the Single Audit Act, which requires annual financial audits of federal award recipients, and GAO's High Risk List, a biennial designation of federal programs vulnerable to fraud and mismanagement [1].

Prior work in anomaly detection has demonstrated the effectiveness of machine learning on imbalanced tabular datasets. Liu et al. introduced Isolation Forest as a tree-based anomaly detection method capable of identifying outliers without defining a normal boundary [2]. Campello et al. extended density-based clustering through HDBSCAN, which eliminates the sensitivity to the epsilon parameter required by traditional DBSCAN and scales efficiently to large datasets [3]. Chen and Guestrin established XGBoost as a state-of-the-art gradient boosting framework particularly well-suited to structured data with class imbalance [4]. Lundberg and Lee introduced SHAP values as a unified framework for interpreting model predictions, enabling feature-level attribution for individual observations [5].

USASpending.gov, maintained by the U.S. Treasury, provides comprehensive publicly accessible data on federal contract transactions [6]. This dataset has been underutilized for systematic risk detection, representing a gap this work directly addresses.

II. METHODOLOGY

A. Dataset Description

The dataset was sourced from USASpending.gov and covers FY2025 contract transactions spanning October 1, 2024 through January 8, 2026 [6]. Three department-level CSV files were obtained corresponding to DoD (agency code 097), DHS (agency code 070), and HHS (agency code 075). After merging, the raw dataset comprised 600,862 contract action rows and 293 attribute columns.

The three departments were selected based on their collective share of federal contracting spend. DoD accounts for approximately 60% of total federal contract obligations,

while DHS and HHS represent significant civilian contracting activity in information technology, professional services, and healthcare procurement - sectors historically associated with elevated procurement risk.

The 293 raw columns encompass financial metrics (obligation amounts, contract ceilings, option values), procurement characteristics (competition type, solicitation procedures, award type), vendor attributes (business size, entity type, country of origin), compliance indicators (cost accounting standards, performance-based service acquisition flags), and socioeconomic designations (HUBZone, veteran-owned, women-owned, 8(a) program participation).

A duplicate check confirmed zero duplicate rows across the merged dataset. A null value analysis revealed that 4 columns were entirely null and were dropped immediately, reducing the column count to 289.

B. Data Cleaning

Data cleaning proceeded through five sequential steps.

1) Column Selection

Following null analysis, 56 risk-relevant feature columns were selected spanning six conceptual risk dimensions: financial risk, procurement risk, compliance risk, vendor risk, performance risk, and socioeconomic fraud risk. Columns with null rates exceeding 50% including executive compensation fields and COVID-19 supplement indicators - were excluded as analytically irrelevant to contract risk detection.

2) Feature Engineering

Two derived features were constructed from raw date columns. First, `days_of_performance` was computed as the difference in days between the period of performance start date and the ordering period end date. Second, `modification_type` was extracted from the modification number field prefix, categorizing contract actions as STANDARD, PRICING, or ADMINISTRATIVE. Five additional ratio features were derived: `obligation_ratio`, `value_gap`, `outlay_ratio`, `scope_expansion_days`, and `solicitation_to_award_days`.

3) Null Row Removal

After column selection, rows containing any null value across the 56 feature columns were dropped. This reduced the working dataset from 600,862 rows to 464,385 rows (77.5% retention). The loss of 136,477 rows was primarily attributable to missing values in `cost_or_pricing_data`, `multi_year_contract`, `contract_financing`, and `days_of_performance` - all of which are analytically significant and could not be imputed without introducing bias into the clustering step.

4) Encoding and Scaling

Forty-one categorical columns were label-encoded using scikit-learn's `LabelEncoder` [7], converting stringvalued fields to integer representations. `StandardScaler` was applied to the full 56-feature matrix,

producing a zero-mean, unit-variance feature space X of shape (464,385, 56) for use in unsupervised clustering [7].

C. Data Preparation - Ground Truth Construction

The absence of human-annotated risk labels required the construction of synthetic ground truth through unsupervised learning. A three-algorithm majority vote framework was developed in which each algorithm independently assigns a binary risk label to each contract. A contract is designated High Risk only when at least two of the three algorithms agree.

1) K-Means Clustering

K-Means clustering with $k=2$ was applied to the standardized feature matrix X [8]. The algorithm was initialized with ten random restarts and converged to a stable partition. Cluster assignment to the High Risk label was determined by comparing mean financial obligation, contract duration, and value gap across clusters. This produced 62,293 flagged contracts (13.4%). A silhouette score of 0.486 confirmed moderate cluster separation.

2) Isolation Forest

Isolation Forest was applied with a contamination parameter of 0.08 and 200 estimators [2]. The contamination parameter was set to 8% based on GAO procurement literature suggesting 5–10% of federal contracts carry some form of irregularity risk [1]. Isolation Forest flagged 37,151 contracts as High Risk (8.0%).

3) HDBSCAN

Hierarchical DBSCAN (HDBSCAN) was selected over standard DBSCAN [3]. Standard DBSCAN requires manual specification of the epsilon neighborhood parameter and has $O(n^2)$ memory complexity, rendering it infeasible on the 464,385-row dataset. HDBSCAN eliminates the epsilon parameter through hierarchical density estimation. Parameters were set to `min_cluster_size=100` and `min_samples=10`. Noise points (`label = -1`) were designated High Risk, yielding 21,172 flagged contracts (4.6%).

4) Majority Vote

The three binary risk vectors were summed to produce a `vote_sum` per contract. Contracts with `vote_sum ≥ 2` were labeled High Risk (1), `vote_sum = 0` were labeled Low Risk (0), and `vote_sum = 1` was designated Uncertain and excluded from classifier training. This produced `df_final` with 436,530 labeled contracts: 39,161 High Risk (8.4%), 397,369 Low Risk (85.6%), and 27,855 Uncertain excluded (6.0%). Among High Risk contracts, 14,439 were unanimously flagged by all three algorithms.

D. Model Creation and Iteration

The supervised classification step utilized the `df_features` dataframe containing 61 columns including additional engineered variables, merged with the `df_final` ground truth labels on index, yielding 436,530 matched rows. Identifier columns were excluded. All remaining categorical columns were label-encoded and residual null values were imputed

with column medians. This produced a final classification feature matrix of shape (436,530, 56).

1) Class Imbalance

The labeled dataset exhibited a class imbalance ratio of 91% Low Risk to 9% High Risk (10.15:1). Two strategies were evaluated: SMOTE [9] and balanced class weights [7]. Cross-validation revealed that SMOTE-trained models exhibited inflated performance metrics due to synthetic observations leaking across training and validation folds — a known limitation of applying SMOTE outside properly nested cross-validation. Accordingly, balanced class weights were selected as the primary strategy.

2) Train-Test Split

The labeled dataset was partitioned into training and test sets using an 80/20 stratified split, preserving the 91/9 class ratio. The training set comprised 349,224 observations and the test set comprised 87,306 observations.

3) Classifier Training

Three classifiers were trained: Logistic Regression (interpretable linear baseline), Random Forest (ensemble of 200 trees), and XGBoost (gradient boosting with 200 estimators). For Logistic Regression and Random Forest, `class_weight='balanced'` was applied. For XGBoost, `scale_pos_weight` was set to 10.15 [4].

4) Cross-Validation

Model generalization was assessed through two protocols: 5-fold Stratified K-Fold cross-validation and a Repeated Stratified K-Fold protocol (5 folds $\times$ 3 repeats = 15 evaluations) applied to the XGBoost model. Both protocols maintained 91/9 class stratification across folds.

5) SHAP Analysis

Post-hoc interpretability was conducted using SHAP TreeExplainer [5], computing exact Shapley values for the XGBoost model on the full test set (87,306 observations), producing a per-observation, per-feature attribution matrix of shape (87,306, 56). Mean absolute SHAP values were aggregated to rank features globally.

III. RESULTS

A. Ground Truth Distribution

The three-algorithm majority vote produced the label distribution shown in Table I across 464,385 clean observations.

TABLE I
Majority Vote Ground Truth Distribution

Label	Count	Pct.	Disposition
HIGH RISK (1)	39,161	8.4%	Training
LOW RISK (0)	397,369	85.6%	Training
UNCERTAIN	27,855	6.0%	Excluded

TOTAL	436,530	100%	Labeled
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Of the 39,161 High Risk contracts, 14,439 (36.9%) were unanimously flagged by all three algorithms, and 24,722 (63.1%) were flagged by exactly two. The 6.0% uncertain rate confirms broad consistency among the three algorithms.

B. Classifier Performance

Table II presents the performance of all three classifiers on the held-out test set of 87,306 contracts using balanced class weights.

TABLE II
Classifier Performance on Test Set ($n=87,306$)

Model	Acc.	Prec.	Rec.	F1	AUC
Log. Reg.	76.0%	19.6%	54.1%	0.29	0.68
Rand. Forest	90.6%	44.4%	17.6%	0.25	0.75
XGBoost*	85.4%	32.8%	59.1%	0.42	0.7574

* Best overall performer across AUC and Recall.

XGBoost outperformed both baselines on the primary oversight-relevant metrics - ROC AUC (0.7574) and Recall (59.1%). Logistic Regression achieved the second-highest recall (54.1%), confirming that the risk signal is partially captured by a linear boundary. Random Forest achieved the highest precision (44.4%) but the lowest recall (17.6%), indicating it defaults to the majority class in borderline cases despite class weighting.

C. Cross-Validation Results

Table III presents fold-by-fold results for the XGBoost 5fold stratified cross-validation.

TABLE III
XGBoost 5-Fold Cross-Validation Results

Fold	Acc.	Prec.	Rec.	F1	AUC
1	85.6%	33.0%	59.0%	0.42	0.7515
2	85.4%	32.5%	58.6%	0.42	0.7585
3	85.3%	32.5%	59.4%	0.42	0.7589
4	85.4%	32.6%	59.2%	0.42	0.7581
5	85.6%	33.1%	59.1%	0.42	0.7598
Mean	85.4%	32.8%	59.1%	0.42	0.7574
Std	± 0.01	± 0.03	± 0.03	± 0.02	± 0.03

The near-zero standard deviations confirm that XGBoost performance is stable and independent of any particular data partition. A Repeated Stratified K-Fold protocol (5 folds $\times$ 3 repeats = 15 evaluations) confirmed AUC stability in the range 0.7515–0.7598 across all evaluations.

D. Confusion Matrix Analysis

On the 87,306-observation test set, XGBoost produced the following confusion matrix outcomes:

- True Positives (TP): 4,739 - High Risk contracts correctly identified (60.5% recall).
- False Negatives (FN): 3,093 - High Risk contracts missed (39.5% missed rate).
- True Negatives (TN): 69,799 - Low Risk contracts correctly cleared (87.8%).
- False Positives (FP): 9,675 - Low Risk contracts incorrectly flagged (12.2%).

In the context of federal contract oversight, false negatives represent the more consequential error type, as they correspond to potentially high-risk awards that escape review. False positives impose a burden of unnecessary review but do not result in undetected risk.

E. SHAP Feature Importance

Table IV presents the top 15 features by mean absolute SHAP value, reflecting each feature's average contribution to the High Risk prediction across all 87,306 test observations.

TABLE IV
Top 15 Features by Mean |SHAP| Value

#	Feature	SHAP	Dimension
1	contract_financing	0.407	Compliance
2	product_or_service_code	0.231	Performance
3	days_of_performance	0.174	Performance
4	total_dollars_obligated	0.159	Financial
5	naics_code	0.153	Performance
6	potential_total_value_of_award	0.118	Financial
7	base_and_all_options_value	0.106	Financial
8	base_and_exercised_options_value	0.098	Financial
9	total_outlayed_amount	0.092	Financial
10	recipient_state_code	0.070	Vendor
11	current_total_value_of_award	0.069	Financial

12	federal_action_obligation	0.066	Financial
13	performance_based_srcv_acq	0.047	Compliance
14	modification_type	0.046	Procurement
15	solicitation_procedures	0.038	Procurement

The dominant feature, contract_financing (SHAP 0.407), captures the type of financing arrangement applied to the contract. Non-standard financing types - including advance payments and progress payments - are associated with elevated audit complexity and have been flagged in GAO oversight reports [1]. Seven of the top fifteen features are financial, confirming that financial structure and magnitude are the primary systematic risk signals.

IV. DISCUSSION AND CONTRIBUTION

A. Interpretation of Results

XGBoost achieved a cross-validated ROC AUC of 0.7574, representing a 51.5% improvement over a random classifier (AUC = 0.50). This confirms that statistically meaningful risk patterns exist within the 56-feature representation of federal contract data and that gradient boosting captures the non-linear decision boundaries separating high-risk from low-risk awards.

The relatively modest precision (32.8%) must be interpreted in context. The ground truth labels were derived from unsupervised clustering on the same feature space used by the classifiers. This creates an inherent limitation: supervised classifiers partially learn to reverse-engineer the clustering logic rather than generalize to an independent ground truth - a known challenge in self-supervised label construction [10].

The recall of 59.1% on the High Risk class is operationally meaningful. In the test set of 7,832 High Risk contracts, the model correctly identifies 4,739 for further review. For oversight agencies operating with limited auditor capacity, a system surfacing priority contracts at 59% recall represents a substantial improvement over the current baseline of unsystematic, sampling-based review.

B. Success - Was the Problem Solved?

The original problem statement identified three gaps: infeasibility of manual review at scale, absence of a unified multi-dimensional risk framework, and lack of automated prioritization. This project addresses all three.

On scale, the pipeline processes all 600,862 FY2025 contracts in a single automated workflow. On multidimensionality, the six-dimension feature framework integrates financial, procurement, compliance, vendor, performance, and socioeconomic risk signals into a single prediction. On prioritization, the output is a traceable list of 39,161 High Risk contracts, each identified by Award ID with SHAP values explaining why it was flagged.

The pipeline constitutes a risk triage tool that prioritizes contracts for human review rather than autonomously determining contractual misconduct - a framing appropriate and intentional given federal audit documentation requirements.

C. Novel Contributions

1) Three-Algorithm Majority Vote for Ground Truth

The primary methodological contribution is the threealgorithm majority vote framework for constructing defensible ground truth labels without human annotation. Prior work has applied single-algorithm approaches to generate pseudo-labels. The use of three structurally different algorithms - a centroid-based method (K-Means), a path-length-based anomaly scorer (Isolation Forest), and a hierarchical density estimator (HDBSCAN) - reduces dependence on any single algorithm's assumptions. The uncertain category excludes observations where only one algorithm flags risk, providing a conservative, highconfidence label set. To the best of the authors' knowledge, this specific combination applied to federal procurement data has not been previously documented.

2) Comprehensive FY2025 Federal Contract Risk Framework

This work presents the first publicly documented end-to-end machine learning pipeline applied to FY2025 USASpending.gov data across DoD, DHS, and HHS simultaneously. The six-dimension feature taxonomy operationalizing 56 features spanning financial, procurement, compliance, vendor, performance, and socioeconomic risk - provides a replicable schema for federal contract risk detection extensible to other agencies and fiscal years.

3) Contract Financing as Dominant Risk Signal

The SHAP analysis produces a novel empirical finding: contract_financing type is the single strongest predictor of High Risk classification (mean $|\text{SHAP}| = 0.407$) - more than 75% higher than the second-ranked feature. This quantitative prioritization has not been previously reported in the federal procurement machine learning literature. The result has actionable implications: automated screening tools should weight financing arrangement type as a primary filter in risk prioritization logic.

4) SHAP-Based Explainability for Procurement Oversight

The application of SHAP TreeExplainer [5] to federal contract risk detection enables contract-level explainability. For any flagged contract, the SHAP decomposition identifies which specific features drove the High Risk prediction and by how much. This property is critical for operational adoption in federal oversight contexts, where auditors must articulate the basis for selecting a contract for review.

D. Future Work

Several directions would strengthen these findings:

- **Human-Annotated Validation:** Collaboration with GAO or agency Inspector General offices to obtain independently labeled contract risk data would enable genuine out-of-sample validation, resolving the selfsupervised circularity limitation.
- **Temporal Generalization:** Training on FY2024 data and evaluating on FY2025 would test whether risk patterns identified in prior years generalize to new contracting cycles.
- **Cross-Agency Deployment:** Extending the pipeline to all 15 federal cabinet departments would assess transferability of the risk model beyond DoD, DHS, and HHS.
- **Hyperparameter Optimization:** Systematic Bayesian optimization (Optuna) of XGBoost hyperparameters would likely improve model performance beyond the current default configuration.
- **Real-Time Integration:** Integration with the USASpending.gov public API would enable nearreal-time risk scoring of newly posted contract awards, shifting oversight from retrospective to proactive.
- **Ensemble of Classifiers:** Combining Logistic Regression (high recall), Random Forest (high precision), and XGBoost (best AUC) through soft voting or stacking may yield a better precision-recall tradeoff.

V. CONCLUSION

This paper presented an end-to-end machine learning pipeline for federal contract risk detection applied to FY2025 USASpending.gov data covering 600,862 contract transactions across DoD, DHS, and HHS. A threealgorithm majority vote framework combining K-Means, Isolation Forest, and HDBSCAN constructed defensible ground truth labels, yielding 39,161 High Risk contracts (8.4%) and 397,369 Low Risk contracts for classifier training.

XGBoost with balanced class weights emerged as the best-performing classifier, achieving cross-validated ROC AUC of 0.7574 (± 0.003) and High Risk recall of 59.1% stable across all 15 cross-validation evaluations. The confusion matrix confirmed that 4,739 High Risk contracts in the test set were correctly identified.

SHAP feature importance identified contract_financing type as the single most influential risk driver (mean $|\text{SHAP}| = 0.407$), followed by product or service code, days of performance, and total dollars obligated. Seven of the top fifteen SHAP features were financial.

The primary contributions are: a novel three-algorithm majority vote framework for unsupervised ground truth construction; a six-dimension, 56-feature federal contract risk taxonomy; a novel empirical finding that contract financing type is the dominant risk predictor; and a SHAPbased explainability framework enabling contract-level audit justification. Together these advance the state of

automated federal contract oversight and provide a replicable methodology for public sector risk detection.

. REFERENCES

- [1] Government Accountability Office, "High-risk series: Federal contract and grant management," U.S. GAO, 2024. [Online]. Available: <https://www.gao.gov/highrisk>
- [2] F. T. Liu, K. M. Ting, and Z.-H. Zhou, "Isolation Forest," in Proc. 8th IEEE Int. Conf. Data Mining (ICDM), 2008, pp. 413–422. doi: 10.1109/ICDM.2008.17
- [3] R. J. G. B. Campello, D. Moulavi, and J. Sander, "Densitybased clustering based on hierarchical density estimates," in Advances in Knowledge Discovery and Data Mining (PAKDD), Lecture Notes in Computer Science, vol. 7819, Springer, 2013, pp. 160–172.
- [4] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in Proc. 22nd ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining, 2016, pp. 785–794. doi: 10.1145/2939672.2939785
- [5] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," Advances in Neural Information Processing Systems, vol. 30, pp. 4765–4774, 2017.
- [6] USASpending.gov, "FY2025 federal contract transaction data - DoD, DHS, HHS," U.S. Department of the Treasury, 2025. [Online]. Available: <https://www.usaspending.gov>
- [7] F. Pedregosa et al., "Scikit-learn: Machine learning in Python," J. Machine Learning Research, vol. 12, pp. 2825–2830, 2011.
- [8] S. P. Lloyd, "Least squares quantization in PCM," IEEE Trans. Inf. Theory, vol. 28, no. 2, pp. 129–137, 1982. doi: 10.1109/TIT.1982.1056489
- [9] N. V. Chawla, K. W. Bowyer, L. O. Hall, and W. P. Kegelmeyer, "SMOTE: Synthetic minority over-sampling technique," J. Artificial Intelligence Research, vol. 16, pp. 321–357, 2002. doi: 10.1613/jair.953
- [10] V. Chandola, A. Banerjee, and V. Kumar, "Anomaly detection: A survey," ACM Computing Surveys, vol. 41, no. 3, pp. 1–58, 2009. doi: 10.1145/1541880.1541882